STELA: Unifying SNOMED CT Logical Expressions and Textual Descriptions for Holistic Concept Representation

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Abstract

SNOMED CT defines clinical concepts through both formal semantic expressions (logical definitions) and human-readable textual descriptions. However, existing embedding methods model these modalities separately: graph-based approaches often overlook lexical nuance, while language models cannot interpret the compositional logic of expressions. This separation prevents a holistic representation of concepts, resulting in embeddings that rely disproportionately on a single modality. We present STELA (SNOMED CT Text and Expression Logical Alignment), an inductive, lightweight, and semantic-aware embedding model that unifies SNOMED CT expressions and textual descriptions in a shared embedding space. Using contrastive learning with minimal parameter updates, the model improves semantic expression-to-text retrieval (MRR 0.81 vs. 0.51 baseline) and generalizes to unseen expressions. This alignment capability is particularly critical for post-coordinated expressions, which lack canonical textual labels. When applied to 23,595 institution-specific post-coordinated expressions, the model demonstrates the capacity for redundancy detection and semantic interoperability. Our approach provides lightweight, semantic-aware embeddings that enhance terminology management and downstream clinical analytics.

Keywords

SNOMED CT, Post-coordinated expression, Clinical concept embedding, Semantic–text alignment, Inductive embeddings, Parameter-efficient fine-tuning, Semantic interoperability

1. Introduction

SNOMED CT is the most comprehensive clinical terminology used internationally, covering disorders, procedures, substances, body structures, and many other domains [1]. Unlike specialized terminologies such as RxNorm, CPT, or ICD-10, SNOMED CT spans multiple clinical areas, and its power lies in its dual representation of clinical meaning: concepts are defined both by human-readable textual descriptions (Fully Specified Names and Synonyms) and formal semantic expressions (description logic) [2]. These two representations may convey complementary aspects of clinical meaning. Ideally, an embedding model should capture both the lexical nuance of the text and the structural precision of the logic.

However, existing clinical concept embedding approaches typically treat these representations in isolation, falling broadly into two categories: graph-based models and language-based models. Embedding models using graph structure, such as translational distance models (e.g., TransE, RotatE) or graph neural networks (GNNs), leverage the ontological topology to learn representations [3, 4]. While effective for capturing hierarchy, these methods suffer from two major limitations in this context. First, they are typically transductive: they assume a fixed set of entities at training time and cannot embed new concepts without retraining or additional approximation steps. Since SNOMED CT continuously

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evolves, transductive models cannot generate embeddings for unseen concepts without full retraining. In addition, purely structural methods often overlook the concepts' lexical information, which is essential for interpreting many concepts. This is particularly problematic for SNOMED CT primitive concepts, which are not fully defined by their logical axioms and rely on textual descriptions to distinguish them from sibling concepts [5]. Models that use only graph structure fail to represent concepts whose meaning comes mainly from their textual labels rather than their logical definitions.

Conversely, text-based embedding models such as BioBERT [6], ClinicalBERT [7], and SapBERT [8] excel at capturing lexical information. However, these models are ill-suited for SNOMED CT expressions, which are not natural language but follow a formal compositional grammar unique to SNOMED CT [9]. While previous work has attempted to parse clinical text into SNOMED expressions [10, 11], learning semantic representations from expressions remains unaddressed.

This disjoint modeling creates a fundamental gap: there is no unified embedding space in which a logical expression can be directly compared to a textual description. This limitation is most acute for concepts that lack one of these representation forms. A prime example is post-coordination, a mechanism that allows users to construct new concepts by combining existing ones to capture fine-grained clinical details not present in the official release. Unlike pre-coordinated concepts, post-coordinated expressions exist only as logical strings, and they possess no permanent ID or human-readable texts, since label generation and translation are costly and offer limited patient retrieval benefit. Because current language models cannot interpret this logic, and graph models are transductive (unable to handle newly added nodes), post-coordinated expressions currently cannot be effectively embedded or compared to standard terminology.

The impact of this gap is significant. At the University Hospitals of Geneva (HUG), we utilize more than 41,000 distinct SNOMED CT concepts and expressions in routine care, with more than half being institution-specific post-coordinated expressions. Without a model that can understand both semantics and human-readable descriptions, these expressions remain isolated from the rest of the clinical terminology, complicating interoperability and downstream analytics.

To address this gap, we develop a lightweight embedding model that aligns SNOMED CT expressions with their corresponding textual descriptions. Using Low-Rank Adaptation (LoRA), we fine-tune SapBERT [8] with only 0.4% additional trainable parameters. In line with SNOMED CT's characteristics, we construct SNOMED-specific alignment pairs that link each concept's expression to its textual description. This approach enables generalization: the model can embed SNOMED CT expressions that were not part of the training set, thereby overcoming the limitations of transductive graph-based methods. We show that the resulting STELA (SNOMED CT Text and Expression Logical Alignment) model achieves strong alignment for both training and unseen concepts, enabling new applications in terminology governance, redundancy detection, and cross-terminology mapping. Figure 1 provides an overview of the approach proposed in this work, summarizing how SNOMED CT data are prepared, how STELA is adapted for semantic–text alignment, and how the resulting embeddings support downstream terminology-governance tasks. Our contributions are threefold: (i) we introduce STELA for aligning SNOMED CT expressions with their textual descriptions while accounting for SNOMED CT's characteristics; (ii) we demonstrate that the model generalizes to expressions not included in the training data; and (iii) through qualitative case studies, we show its potential for terminology-management tasks such as redundancy detection and cross-terminology mapping.

2. Background

SNOMED CT concepts and their representation forms SNOMED CT is organized into hierarchies of concepts representing clinical meanings. Each concept is comprised of three distinct representation forms: a unique concept identifier (SCTID), a set of textual representation (descriptions), and a semantic representation (description logic). Figure 2 illustrates the different representation forms.

Descriptions link the concept to human-readable terms [12]. They include the Fully Specified Name (FSN), which uniquely identifies and disambiguates the concept meaning and ends with a semantic tag

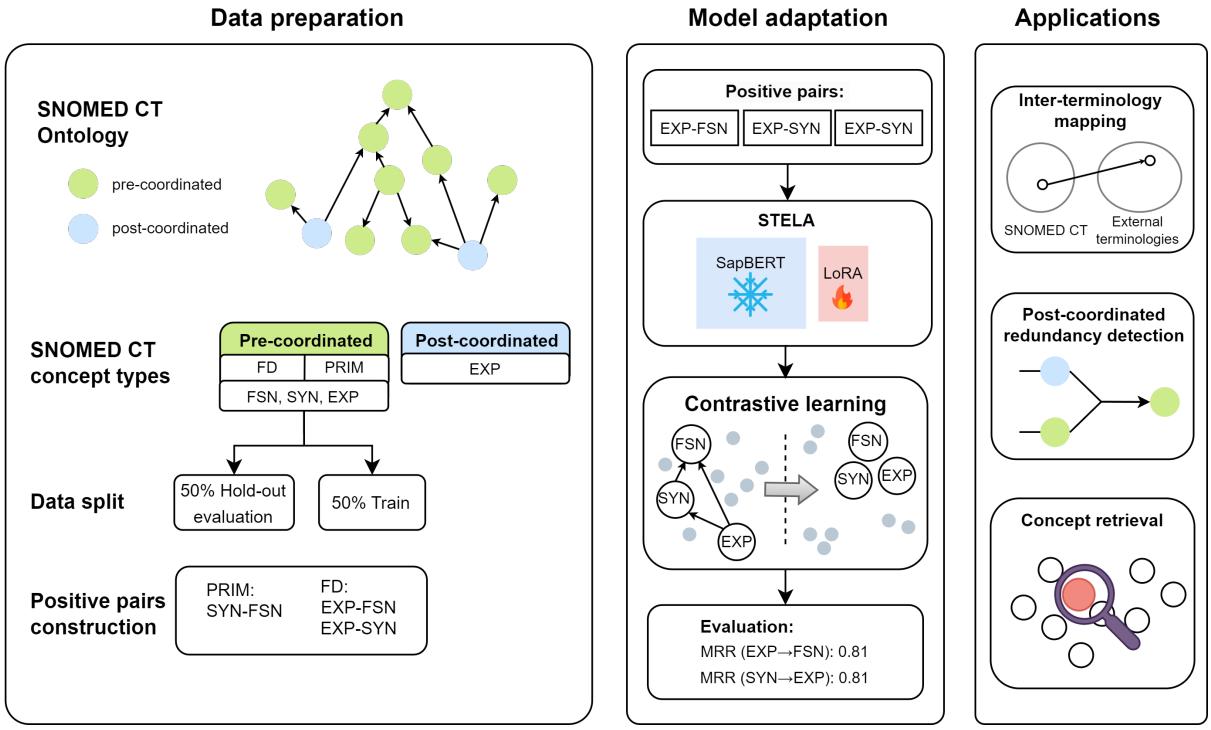


Figure 1: Workflow of STELA for aligning SNOMED CT expressions with their textual forms and enabling downstream applications. FD: fully defined concepts; PRIM: primitive concepts; EXP: expression; FSN: fully specified name; SYN: synonym.

indicating its domain, as well as synonyms (SYN), which are alternative terms used in clinical practice. Every concept must have exactly one unique FSN, but may have multiple synonyms. SNOMED CT is also a semantic network in which the concepts are interconnected. Thus, beyond textual representation, each concept also possesses a logical definition, referred to as expression (EXP). The EXP of a concept is inferred from its structured combination of defining relationships to other concepts [13]. The syntax used to build expressions is SNOMED CT compositional grammar [14].

Pre-coordination and Post-coordination SNOMED CT concepts can be categorized into two distinct types: pre-coordinated concepts and post-coordinated expressions. Figure 2 summarizes different concept types.

Pre-coordinated concepts are those explicitly distributed in the official SNOMED CT release. They possess a SCTID, a FSN, one or many SYN, and an EXP. These concepts are further categorized by their definition status:

- **Fully Defined (FD):** The logical definition of the concept is *both necessary and sufficient*. The logic alone is distinct enough to uniquely differentiate the concept from all others in the hierarchy [15].
- **Primitive (PRIM):** The concept's logical definition is *necessary but not sufficient*. The logic describes the concept's attributes, but is not unique enough to distinguish itself from its parents or siblings without the explicit human-readable name. Thus, it relies on the lexical description for full disambiguation [5]. As shown in Figure 2, the concept *Atypical appendicitis* is considered primitive, as its logic definition is not sufficient to distinguish *atypical appendicitis* from any other type of *appendicitis*.

Post-coordinated expressions are utilized when a specific clinical idea is required but absent from the official release. To maintain clinical granularity without waiting for a new official release, institutions may generate a new representation using SNOMED CT compositional grammar. The post-coordinated

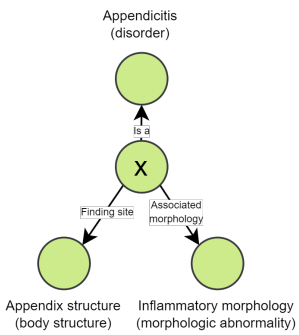
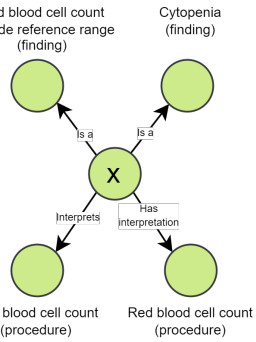
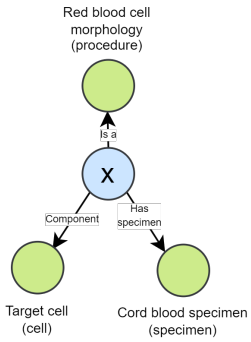
	Pre-coordinated		Post-coordinated
	PRIM	FD	
Definition in ontology			
FSN	Atypical appendicitis (disorder)	Red blood cell count below reference range (finding)	-
SYN	Atypical appendicitis	RBC count low, Erythropenia, Red blood cell count low, etc...	-
EXP	|Appendicitis (disorder)| : {|Finding site (attribute)| = |Appendix structure (body structure)|, |Associated morphology (attribute)| = |Inflammatory morphology (morphologic abnormality)| }	|Cytopenia (finding)| + |Red blood cell count outside reference range (finding)| : { |Has interpretation (attribute)| = |Below reference range (qualifier value)|, |Interprets (attribute)| = |Red blood cell count (procedure)| }	|Red blood cell morphology (procedure)|: {|Has specimen (attribute)|= |Cord blood specimen (specimen)|} {|Component (attribute)|= |Target cell (cell)|}

Figure 2: Comparison of concept types and their representation forms. Concept identifiers are excluded from expressions for readability. Concept types are distinguished by color: pre-coordinated concepts (green) and post-coordinated expressions (blue). The focal concept of each example is annotated with "x". FSN: Fully Specified Name; SYN: Synonym; EXP: Expression; PRIM: Primitive concept; FD: Fully Defined concept.

expression is a logical composition of two or more existing identifiers (e.g., combining a disorder concept with a specific severity concept)[16]. Unlike pre-coordinated concepts, post-coordinated expressions exist only as logical strings, they do not possess a permanent SCTID, FSN, or SYN.

Text-based clinical concept embedding models and limitations Clinical concept embedding is the process of translating discrete medical concepts, such as diagnosis codes (ICD-10), medications (RxNorm), or procedure codes (CPT) into continuous, low-dimensional vector representations. Current state-of-the-art approaches rely on pre-trained language models to encode human-readable descriptions found in medical ontologies or clinical notes. To evaluate their performance, we used all SNOMED CT concepts in the January 2025 official release and performed the textual descriptions alignment between concepts' FSN and SYN. As shown in Table 1, SapBERT [8] achieves the highest Mean Reciprocal Rank (MRR) of 0.72 on the standard task of retrieving SNOMED CT concepts' FSN given SYN. Here, MRR quantifies how highly the correct counterpart appears in a ranked list by averaging the inverse of its rank, the closer the MRR is to 1, the better the model's alignment performance. This superior performance is attributed to SapBERT's self-alignment training on the UMLS Metathesaurus, which explicitly optimizes the model to cluster synonymous medical strings closer together in embedding space.

However, while SapBERT excels at processing natural language descriptions, it struggles to generalize to SNOMED CT EXPs. EXPs are not natural sentences, but rather structured strings governed by a strict compositional grammar that combines nested concept descriptions with logical operators, as shown in Figure 2. Our preliminary experiments on FD concepts (Table 2) reveal that SapBERT struggles to interpret this syntax. When embedding raw expressions containing logical symbols (e.g., +, :, =, {, |, }), the alignment performance drops significantly compared to standard text inputs. Notably, removing the logical symbols actually improves performance (MRR increases from 0.51 to 0.60). This indicates that the model treats the logical syntax as "noise" or confusing punctuation rather than using it to resolve the semantic relationship between the concepts. Even with symbols removed, the performance remains

Table 1
Comparison of embedding models on SNOMED CT description alignment (SYN–FSN). SapBERT outperforms both generic and clinical BERT variants.

Model	MRR
all_MiniLM	0.57
e5_base	0.64
SapBERT	**0.72**
gte_base	0.66
BioClinical_BERT	0.13
Clinical_BERT	0.39

Table 2
Performance of SapBERT on aligning SNOMED CT expressions to their textual descriptions of FD concepts. *With symbols* retains the standard compositional grammar, whereas *Without symbols* removes all logical operators. In both settings, SCTID identifiers are stripped from expressions before embedding.

Expression Input	Task	MRR	Mean Rank
With symbols	EXP-FSN	0.51	165.53
Without symbols	EXP-FSN	**0.60**	**127.88**

below the benchmark for standard textual descriptions, underscoring the inability of current text-based encoders to capture the semantics of compositional clinical logic.

Parameter-Efficient Fine-Tuning (PEFT) Fine-tuning language models on domain-specific data typically requires updating all model parameters, a process that is computationally expensive and prone to catastrophic forgetting, where the model loses its pre-trained general knowledge. For clinical applications, where computational resources and the amount of data are often limited, full fine-tuning is often impractical[17]. Parameter-Efficient Fine-Tuning (PEFT) addresses this by freezing the pre-trained backbone and updating only a small set of additional parameters. In this work, we utilize Low-Rank Adaptation (LoRA) [18]. LoRA injects trainable rank decomposition matrices into the transformer layers while keeping the original weights frozen. This approach allows the model to adapt to the specific syntax of SNOMED CT EXP while preserving the rich biomedical knowledge embedded in the original model weights.

Contrastive Learning and Loss Functions Contrastive learning is a representation learning paradigm that aims to construct an embedding space where similar data samples are pulled closer together, while dissimilar pairs are pushed apart [19]. This approach is particularly effective for cross-modal alignment, where the objective is to learn a shared embedding space for heterogeneous inputs.

A fundamental challenge in contrastive learning is the selection of informative training pairs. Traditional approaches, such as triplet loss [20], rely on inputs organized into specific (anchor, positive, negative) triplets. However, this method is highly sensitive to the sampling strategy: "easy" triplets contribute vanishingly small gradients, while overly "hard" triplets can lead to training instability. Consequently, these methods often require complex, heuristic-based mining stages to select effective negatives.

Wang et al. [21] addressed this by unifying the pair mining and weighting processes into a single framework. Rather than discarding uninformative pairs through a discrete selection step, MultiSimilarity Loss (MS Loss) operates on the full batch and applies General Pair Weighting (GPW). It dynamically assigns weights to every pair based on their relative similarity:

1. Self-similarity: The absolute cosine similarity of the pair itself.
2. Relative similarity: The pair's similarity relative to other positive or negative pairs in the batch.

Table 3
Types and counts of positive training pairs used for STELA fine-tuning.

concept type	Anchor	Positive	Count	Training role
FD	EXP	SYN	71,256	EXP alignment
FD	EXP	FSN	72,785	EXP alignment
PRIM	SYN	FSN	141,524	Update new SYN-FSN

MS Loss automatically assigns higher importance to the most informative positives and negatives. This enables models to learn fine-grained distinctions in dense embedding spaces without the need for manual negative sampling strategies.

3. Methods

3.1. Data preparation

We used the January 2025 release of the international version of SNOMED CT, which contains 370,932 pre-coordinated concepts (145,585 FD concepts and 225,347 PRIM concepts). For model development, we randomly sampled approximately half of the FD concepts (n = 72,792) and PRIM concepts (n = 124,470) for the training set, the remaining concepts were held out as an unseen test set to evaluate the model's generalization capacity.

Before constructing training pairs, all EXP were preprocessed: concept identifiers (SCTIDs) were removed, but both the human-readable terms and logical symbols were retained. SYN were filtered to remove redundancy: if a SYN was identical to the FSN once the semantic tag was stripped from the latter, it was discarded. This ensured that training pairs contained only distinct lexical variants. Positive training pairs were constructed from two of the three representation forms (EXP, FSN, SYN), as summarized in Table 3. For FD concepts, we generated semantic–text pairs (EXP–FSN, EXP–SYN). For PRIM concepts, we focused on SYN–FSN pairs. This strategy served a dual purpose: first, it refreshed SapBERT's core alignment capabilities on the latest 2025 SNOMED CT release, mitigating catastrophic forgetting. Second, it provided a robust signal for primitive concepts, whose logical expressions are often less semantically rich than their textual descriptions. The number of positive training pairs and their training roles are summarized in Table 3.

3.2. Backbone model selection and fine-tuning

We divided the model development process into three stages: backbone selection, parameter-efficient fine-tuning, and the training objective.

To select a backbone embedding model, we first performed a zero-shot SYN-FSN retrieval task using all pre-coordinated concepts. As shown in Table 1, SapBERT achieved the best alignment performance among candidate embedding models, making it the most suitable baseline. We then adapted SapBERT using LoRA[18]. This method freezes the original 110M SapBERT parameters and injects small, trainable low-rank matrices into the attention layers. This approach allowed us to train only 0.4% of the total parameters. For training, we followed SapBERT methodology and employed the Multi-Similarity Loss [21] with cosine similarity as the distance metric. To improve training efficiency, we applied online hard negative mining via the Multi-Similarity Miner [21], which selects the most informative samples within each mini-batch. All training and optimization hyperparameters are detailed in Table 4. The full fine-tuning run completed in approximately 152 minutes on a single NVIDIA GeForce RTX 3060 GPU.

3.3. Evaluation tasks

We evaluated STELA across three retrieval tasks designed to quantify the alignment quality between a concept's different representation modalities:

Table 4
Key hyperparameters for STELA fine-tuning.

Parameter	Value
LoRA	
Rank (r)	8
Alpha (α)	16
Dropout	0.1
Target Modules	Query, Key, Value
Loss Function	Multi-Similarity Loss
MS Loss α	2.0
MS Loss β	50.0
MS Loss base	0.5
Online hard negative mining	Multi-Similarity Miner
Epsilon (ϵ)	0.2
Training and Optimization	
Optimizer	AdamW
Learning Rate	1e-5
Weight Decay	0.01
Batch Size	46
Max Sequence Length	96 tokens
Epochs	3
Pooling	*[CLS]*
Hardware and Precision	
GPU	NVIDIA GeForce RTX 3060
Mixed Precision	FP16 (with FP32 weights)

1. **EXP–FSN retrieval (FD concepts)**: Retrieving the FSN given a concept's EXP.
2. **SYN-EXP retrieval (FD concepts)**: Retrieving the EXP given a concept's SYN.
3. **SYN–FSN retrieval (all pre-coordinated concepts)**: Retrieving the FSN given a concept's SYN.

For each task, retrieval was performed by computing pairwise cosine similarities between L2-normalized embeddings of the two modalities (e.g., EXP vs. FSN). For every query representation, all candidate counterparts were ranked in descending order of cosine similarity, and the position of the true counterpart within this ranked list was recorded. A retrieval was considered correct if the top-ranked item corresponded to the same concept.

The first two tasks assess the model's ability to map between a concept's semantic (EXP) and lexical (FSN/SYN) forms, while the third task evaluates the model's alignment capacity on purely lexical synonym pairs. Evaluation was restricted to pre-coordinated concepts. Furthermore, since primitive concepts' EXP are not semantically discriminative, EXP-based retrieval was applied only to FD concepts.

Performance was quantified using Mean Reciprocal Rank (MRR), mean rank, and Top-K accuracy ($K = 1$–10). Higher MRR values indicate better performance, with values closer to 1 reflecting that the correct item is retrieved among the top-ranked candidates. Mean rank reports the average position of the correct item in the ranked list, with lower values indicating that the model places the correct match closer to the top. Top-K accuracy measures the proportion of queries for which the correct counterpart appears within the first K retrieved items, providing a practical view of retrieval quality at different cutoff levels. For all tasks, embeddings for each representation form were L2-normalized. Retrieval was evaluated on both all concepts and the held-out unseen concepts to assess inductive generalization.

4. Results

We evaluated our fine-tuned STELA model against the SapBERT baseline on the three retrieval tasks. A detailed comparison of retrieval task performance, based on MRR and mean rank, is presented in Table 5. The corresponding Top-K accuracy for each task is plotted in Figure 3. Our model demonstrated a substantial improvement in aligning expressions while fully preserving the baseline's state-of-the-art performance on text-only alignment.

EXP–FSN retrieval showed the largest improvement over the SapBERT baseline. Across all concepts, our model achieved an MRR of 0.81 compared to 0.51 for the baseline, with a mean rank of 12.33 versus 165.53. For unseen concepts held out during training, the model's performance was equally high, confirming its strong generalizability (MRR 0.80 vs. 0.5, mean rank 7.71 vs. 156.41).

SYN-EXP retrieval followed the same trend. The fine-tuned model substantially outperformed the baseline (MRR 0.81 vs. 0.53; mean rank 95.80 vs. 206.35). On unseen concepts, improvements remained consistent (MRR 0.81 vs. 0.53; mean rank 104.09 vs. 211.75). These improvements hold both for concepts seen during training and for the held-out unseen set, confirming that the model generalizes beyond memorized pairs.

For SYN–FSN retrieval, performance was slightly improved compared to the baseline (MRR 0.78 vs. 0.72; mean rank 163.16 vs. 179.98). This was expected, as SYN–FSN positive pairs were explicitly included during training for synonym updating, and LoRA adaptation mitigates catastrophic forgetting of SapBERT's original synonym alignment capacity.

As shown in Figure 3, the plots for EXP-FSN and SYN-EXP clearly illustrate that our fine-tuned model (blue line) consistently outperforms the baseline (orange line) across all K values. However, for SYN-FSN, the plot reflects the findings in Table 5, the improvement is primarily observed at K=1, while for larger K values, the performance of both models is effectively identical.

Table 5
Model performance across all tasks is measured by Mean Reciprocal Rank (MRR) and Mean Rank. "Unseen" indicates evaluation on held-out test concepts. All retrievals are performed over the entire SNOMED CT concept set.

	EXP–FSN		**SYN-EXP**		**SYN–FSN**	
Model	All	Unseen	All	Unseen	All	Unseen
Mean Reciprocal Rank (MRR)						
SapBERT	0.51	0.50	0.53	0.53	0.72	0.72
STELA (Ours)	**0.81**	**0.80**	**0.81**	**0.81**	**0.78**	**0.78**
Mean Rank						
SapBERT	165.53	156.41	206.35	211.75	179.98	188.19
STELA (Ours)	**12.33**	**7.71**	**95.80**	**104.09**	**163.15**	**173.14**

5. Discussion

5.1. Expression alignment and generalization

Our results demonstrate that the STELA model successfully learns to interpret the structured syntax of SNOMED CT expressions, a task for which the original SapBERT was not trained. This is shown by the performance gains in both the EXP–FSN and SYN-EXP retrieval tasks (Table 5). Compared to the baseline SapBERT, our model achieves both high MRR (e.g., 0.81 for EXP–FSN) and low mean ranks (e.g., 12.33). This combination indicates that the model's interpretation of expressions is not only accurate for the top-ranked results but also substantially more consistent, avoiding the extreme mis-rankings observed in the baseline (mean rank 165.53).

Crucially, these gains were achieved while slightly improving the state-of-the-art performance of

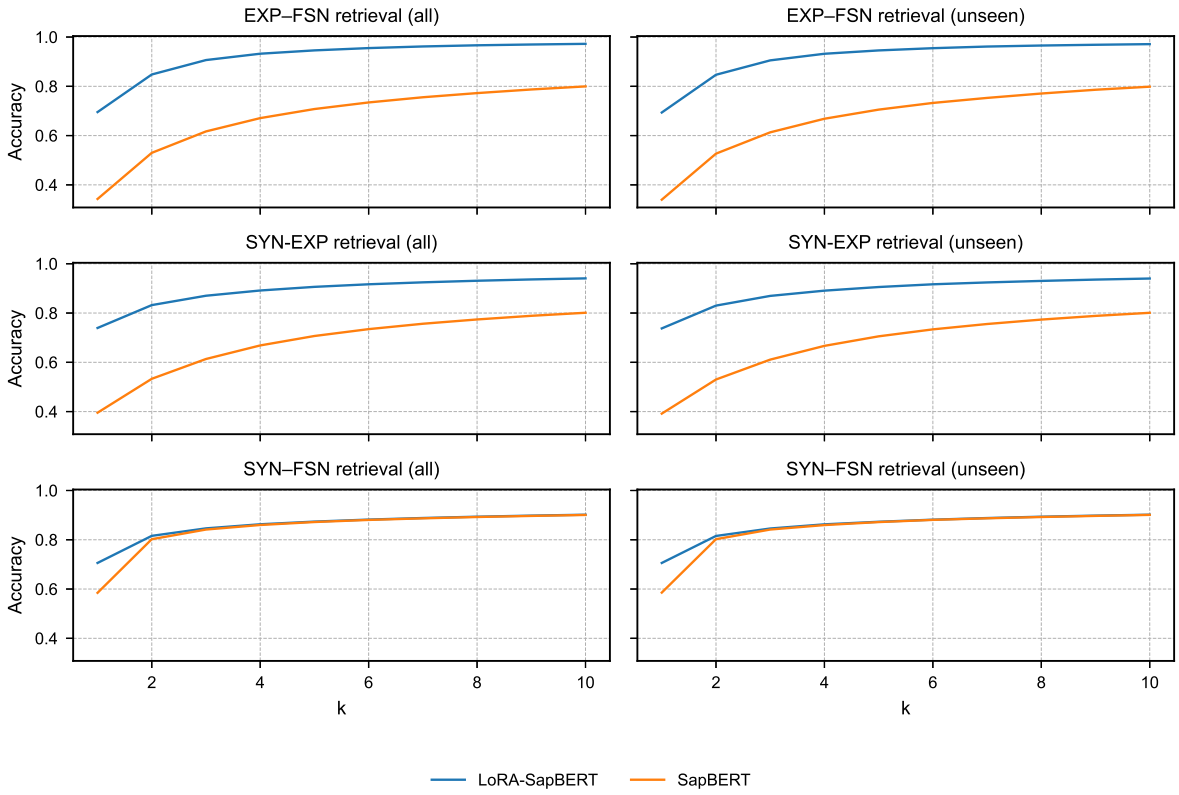


Figure 3: Top-K accuracy across retrieval tasks.

the baseline on the text-only SYN–FSN retrieval task. This confirms that our fine-tuning approach successfully mitigated catastrophic forgetting and updated the model with SNOMED CT recent version.

We do observe, however, that the mean rank for the SYN–FSN task remains high (approx. 163-173), even while the MRR is high. This suggests that the SNOMED CT synonym lexicon itself contains a small minority of highly ambiguous or difficult synonyms (e.g., generic, single-word synonyms, or abbreviations) that are ranked very poorly. These few outlier failures negatively impact the mean rank metric, while the MRR, which is less sensitive to such outliers, confirms that performance for the vast majority of queries is excellent.

Finally, the model's performance on the unseen concepts held out during the training phase was nearly identical to its performance on the full set (e.g., MRR 0.80 on unseen EXP–FSN vs. 0.81 on all). This demonstrates STELA's strong generalization capacity. Rather than relying on memorization of the training expressions, the model has instead learned the underlying grammatical structure of SNOMED CT. This inductivity is a critical feature, making the model useful for an evolving ontology like SNOMED CT.

5.2. Expression–text semantic gap analysis and choice of representation form for embedding

In constructing positive pairs, we applied different strategies depending on concept status: EXP–FSN and EXP–SYN pairs were used for FD concepts, whereas PRIM concepts were limited to SYN–FSN pairs. This design follows the SNOMED CT principle that the EXP of a PRIM concept is often not semantically representative of its textual description (for a concrete example, see Figure 2).

Figure 4 shows the resulting semantic (EXP) - textual (FSN and SYN) description similarities by concept status. FD concepts exhibit very high EXP–FSN and EXP–SYN similarity (median $\approx$ 0.90 and 0.87). This performance is expected, given that the model was explicitly trained on these pairs. Furthermore, FD concepts inherently possess a higher degree of semantic alignment between their logical definitions

and textual labels. The tight distributions confirm that the model learned the alignment task effectively. Moreover, trained (green) and unseen (red) concepts follow nearly identical distributions, demonstrating strong inductive behavior.

In contrast, PRIM concepts display substantially lower EXP–FSN and EXP–SYN similarities (median $\approx$ 0.70 and 0.68) with wider variance. This reflects the inherent incompleteness of EXP for PRIM concepts. This aligns with SNOMED CT's characterization of PRIM concepts as not sufficiently defined by their logical axioms[5].

Finally, EXP–SYN similarities exhibit higher variability than EXP–FSN for both PRIM and FD concepts, indicating that SYN often provides a weaker or less precise representation of a concept's intended semantic meaning. For instance, for *Antibiotic enterocolitis (disorder)* (SCTID: 397696004), the EXP aligns well with the FSN, whereas its SYN *Janbon's syndrome* does not capture the underlying semantics. Thus, the lower EXP–SYN similarity observed in such cases reinforces that the FSN provides a more accurate semantic match to the EXP. This variability in SYN quality also helps explain the large mean rank observed in the SYN–FSN retrieval task shown in Table 5: a small subset of weak or ambiguous SYN yields extremely poor retrieval positions, and these outlier cases substantially inflate the mean rank despite high overall MRR.

A practical implication of these findings is that EXP-based embeddings are reliable only when the EXP contains sufficient semantic detail. This is typically the case for FD concepts and for well-built post-coordinated EXP, which lack textual descriptions and rely entirely on their compositional semantics. Conversely, for pre-coordinated PRIM concepts, where the semantic logic is incomplete and not discriminative, the FSN should be used as the representation form for embedding.

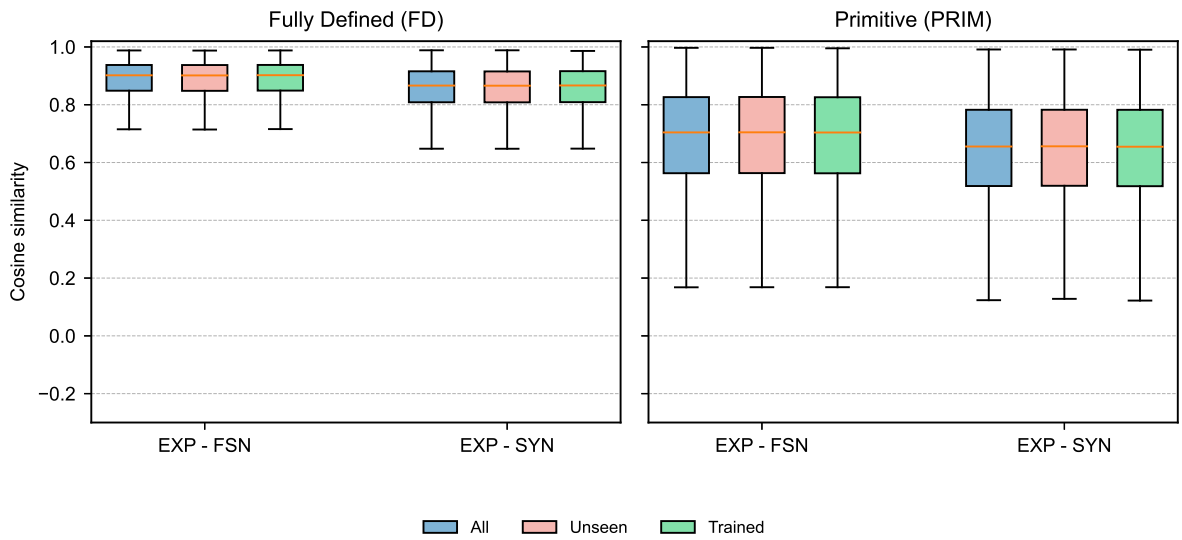


Figure 4: Distribution of cosine distances between concept modalities (EXP, FSN, SYN) by concept status (FD vs. PRIM). Similarities are shown for all concepts, the training subset, and the held-out evaluation subset. Higher values on the y-axis indicate higher semantic-textual description similarity of the same concept.

5.3. Qualitative analysis of post-coordinated expressions

Beyond quantitative metrics, the primary value of our inductive model lies in its ability to analyze real-world, institution-specific post-coordinated EXP that lack any FSN or SYN. To demonstrate this, we embedded all 23,595 post-coordinations EXP used at HUG using STELA and retrieved for each the nearest neighbor from a combined corpus of textual descriptions from SNOMED CT pre-coordinated concepts and other UMLS terminologies (e.g., LOINC, ATC). To assess whether cosine similarity reflects clinically meaningful correspondence, a medical expert evaluated a stratified sample of 50 post-coordinations and their best-matched UMLS concepts, randomly selecting 10 examples from each 0.1 similarity interval

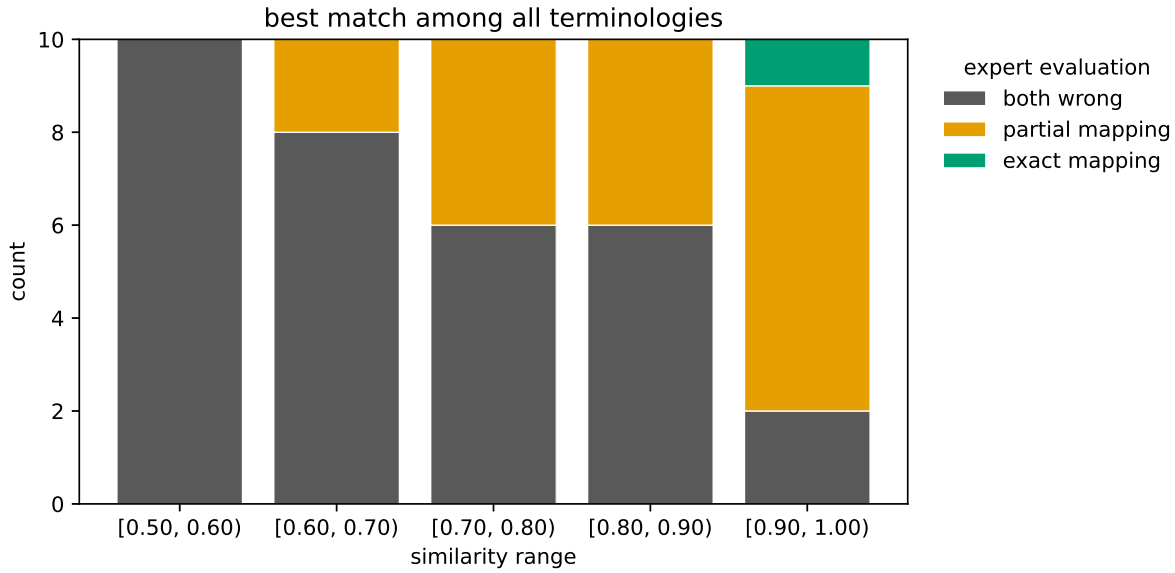


Figure 5: Expert evaluation across similarity ranges, evaluated on 50 post-coordination samples

between 0.50 and 1.00. The result (Figure 5) shows a clear trend: higher cosine similarity values corresponded to a greater proportion of correct or partially correct mappings. Low-similarity intervals were dominated by incorrect matches. This confirms that the embedding space captures clinically coherent structure and that cosine similarity can be interpreted as a proxy for mapping confidence. Examination of the highest-similarity top-ranked matches revealed two principal use cases for STELA when embedding post-coordinated EXP:

1. **Redundancy detection for terminology management.** When the nearest neighbor of a post-coordinated EXP was an existing SNOMED CT pre-coordinated concept, this may suggest redundancy or overlap between institutional post-coordinated EXP and the official terminology. For example, the post-coordination "|*Pancreaticoduodenectomy*| + |*Surgery using robotic assistance*|" was correctly aligned with the pre-coordinated concept *Pancreaticoduodenectomy using robotic assistance* (SCTID: 1260293004), which appeared in a later SNOMED CT release. This capability provides terminology managers with a practical tool for identifying redundant institutional post-coordinations and harmonizing them with concepts that have since been incorporated into the official release.
2. **Semantic interoperability and cross-terminology mapping.** Post-coordinated EXP often encode clinically relevant details that are not represented in the current SNOMED CT release. When such enriched EXP are present, STELA can be used to map them to equivalent or more specific concepts in external terminologies, thereby facilitating interoperability across institutions that may use different coding systems. For example, the expression "|*Beclometasone*| + |*Formoterol*| + |*Glycopyrronium*|" specifies a triple-substance combination that SNOMED CT is not currently available, yet the model correctly aligned it with its precise counterpart in the ATC terminology (ATC code: R03AL09). Likewise, the post-coordination EXP "|Platelet count|: |Has specimen| = |Cord blood specimen|" was mapped to the LOINC concept *Platelets [#/volume] in Cord blood by Estimate* (LONIC code: 55802-3), which is more specific than any single SNOMED concept deviated from *Platelet count* (SCTID: 61928009). These examples illustrate the model's potential to act as a 'semantic bridge', enabling interoperability by mapping enriched, context-specific SNOMED CT EXP to their precise counterparts in other standard vocabularies.

This qualitative analysis confirms our model's practical utility beyond quantitative benchmarks. It functions as a powerful tool for real-world terminology management, enabling both the identification of redundant institutional concepts and the creation of a semantic bridge to external vocabularies, directly

enhancing clinical data interoperability.

5.4. Limitations and Future Work

Despite strong retrieval performance, this study has several limitations that open avenues for future work. First, our qualitative evaluation of post-coordination, while promising, was not based on a public "gold standard" benchmark, as one does not currently exist. Second, we did not evaluate our embeddings in downstream clinical prediction tasks, such as patient mortality or readmission modeling. Furthermore, our analysis was confined to expressions from a single institution and a monolingual (English) scope. Finally, the embedding space produced by STELA does not support vector arithmetic in the sense commonly associated with classical word embeddings (e.g., $e(x) + e(y) \neq e(x + y)$). Preliminary inspection indicates that the model does not encode compositional semantics in an additive manner. This is likely a consequence of using [CLS] pooling.

Future work can address these gaps and build directly on our findings. The model's ability to map EXP to pre-coordinated concepts is immediately applicable to ontology concept management. For example, integrating the embeddings into a vector database could provide real-time semantic search and automated redundancy detection. This directly supports scaling up our qualitative findings on inter-terminology mapping, formalizing the model's role as a "semantic bridge" to external terminologies such as LOINC and ATC, to enhance system-wide interoperability. Finally, the model's understanding of SNOMED CT compositional grammar could support the development of a novel, semantic-aware tokenizer for patient data modeling. Such a tokenizer would address a common limitation in EHR-based self-supervised learning, where many studies must reduce vocabulary size by discarding rare concepts or replacing them with parent concepts to maintain a manageable dictionary, as noted in [22]. These strategies lead to a substantial loss of clinical granularity. In contrast, the inductive capabilities of our embedding model could enable a tokenizer that generates meaningful representations for rare and post-coordinated concepts on the fly, allowing downstream predictive models to exploit the full semantic richness of clinical data without facing an unmanageably large vocabulary.

6. Conclusion

In this work, we introduce STELA, a parameter-efficient adaptation of SapBERT designed to align SNOMED CT's semantic representation with its corresponding textual descriptions. SapBERT already provides a strong backbone thanks to its UMLS pretraining and its exposure to SNOMED CT labels. As a result, a LoRA fine-tuning step is sufficient. By constructing positive pairs that respect SNOMED CT characteristics and fine-tuning the backbone with only 0.4% additional parameters, we enable the model to interpret the compositional syntax of expressions while fully preserving its strong synonym alignment capabilities. Our quantitative results demonstrate substantial improvements in expression-text retrieval tasks and, crucially, consistent performance on unseen concepts, confirming the inductive nature of our approach. This enables the embedding of post-coordinated expressions into the same unified embedding space as pre-coordinated concepts, despite the absence of their textual representation. Qualitative analyses on real institutional data further highlight the model's practical value for terminology governance such as redundancy detection and cross-terminology semantic mapping.

Overall, our findings show that lightweight alignment provides an effective and scalable solution for integrating SNOMED CT expressions into modern embedding workflows. This work lays the foundation for future efforts such as semantic-aware tokenization, semantic search in vector databases, and downstream patient-level modeling that leverages the full granularity of SNOMED CT.

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Declaration on Generative AI

During the preparation of this work, the authors used **ChatGPT**, **Gemini**, and **Grammarly** in order to: **Grammar for spelling check**, **Paraphrase and reword**, and **Improve writing style**. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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